MECH321 Dynamics of Engineering Systems

Lecture 3
Transfer Functions

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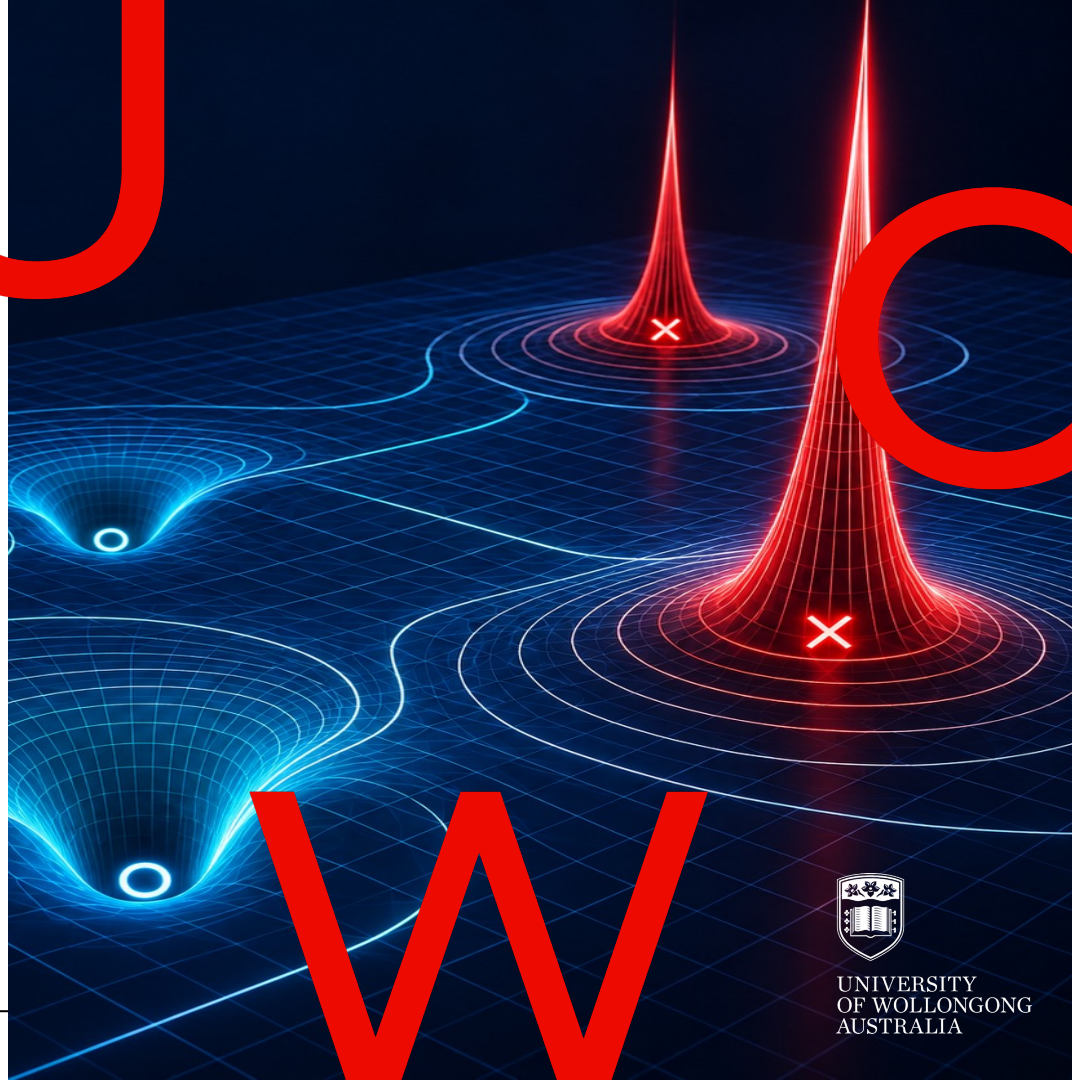


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PART 1

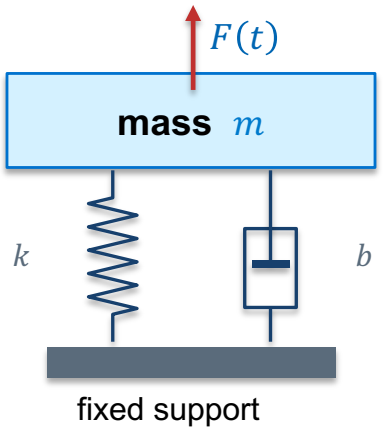
Transfer functions from ODEs

01 ODE to transfer function

02 Zero-state meaning

03 Input families and internal-node elimination

Fixed-base system: input and initial-state response



Measure $x(t)$ upward from static equilibrium. Gravity cancels in this coordinate.

New question

Lecture 1: the established model

Input: force $F(t)$ Output: upward displacement $x(t)$

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

Initial state: $x_0 = x(0^-), \quad v_0 = \dot{x}(0^-)$ not necessarily zero.

Lecture 2: transformed equation, including initial state

$$(ms^2 + bs + k)X(s) = F(s) + (ms + b)x_0 + mv_0$$

How do the applied force and the initial state each contribute to $x(t)$?

Total response: zero-state plus zero-input

CORE

$$X(s) = X_1(s) + X_2(s), \qquad D(s) = ms^2 + bs + k$$

Zero-state response $X_1(s)$

$x_0 = v_0 = 0$ Keep the applied force $F(t)$.

$$X_1(s) = \frac{F(s)}{D(s)}$$

Response driven by the applied input.
Includes transient and sustained components.

A stable system may settle at nonzero displacement.

Zero-input response $X_2(s)$

$F(t) = 0$ Keep the initial state.

$$X_2(s) = \frac{(ms + b)x_0 + mv_0}{D(s)}$$

Free response driven by the initial state.

With m, b, k > 0, damping dissipates initial energy. **Undamped motion may persist.** Unstable motion may grow.

The transfer function

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

Unit impulse: input transform 1; output transform G(s)

$$X_1(s) = G(s)F(s)$$

Definition: the output-to-input ratio in the s-domain, with $x_0 = v_0 = 0$

Limitations

$G(s)$ alone **does not include the initial-state response**.
For nonzero x_0 or v_0, add $X_2(s)$ to $G(s)F(s)$.
It does not specify a complete set of internal states.
Lecture 6 introduces a state-space representation.

DC gain: the constant-input ratio

$K_{DC} = G(0) = \frac{1}{k}$

For a stable plant, final output = DC gain × constant input.
Here its units are displacement per force.

Four useful input shapes

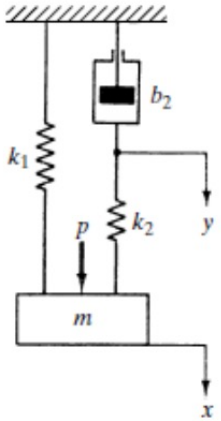
Input	What it tests	Time signal	Laplace transform
Impulse	Natural response after a short kick	$f(t) = \delta(t)$	$F(s) = 1$
Step	New equilibrium and transient overshoot	$f(t) = 1$	$F(s) = \frac{1}{s}$
Ramp	Tracking an input that keeps moving	$f(t) = t$	$F(s) = \frac{1}{s^2}$
Sinusoid	Response at one frequency	$f(t) = \sin(\omega t)$	$F(s) = \frac{\omega}{s^2 + \omega^2}$

Unit inputs, applied from $t = 0$. Signals are zero for $t < 0$.

Mechanical model with a massless junction

EXAMPLE

Question Find the applied-force-to-displacement transfer function.



Downward is positive for the mass and junction. Both displacements are deviations from static equilibrium. Assume linear springs, a viscous damper and zero initial conditions.

1 Mass force balance

$$m\ddot{x} + k_1 x + k_2(x - y) = p(t)$$

2 Massless junction: spring force equals damper force

$$0 = k_2(x - y) - b_2\dot{y}$$

The junction moves without inertia; its net force is zero.

Internal-node elimination

EXAMPLE

3 Transform both balances and eliminate the internal displacement

$$(ms^2 + k_1 + k_2)X(s) - k_2 Y(s) = P(s)$$

$$Y(s) = \frac{k_2}{b_2 s + k_2} X(s)$$

Eliminate Y(s) using the junction balance.

$$\left(ms^2 + k_1 + k_2 - \frac{k_2^2}{b_2 s + k_2}\right) X(s) = P(s)$$

$$[(ms^2 + k_1 + k_2)(b_2 s + k_2) - k_2^2] X(s) = (b_2 s + k_2) P(s)$$

$$\frac{X(s)}{P(s)} = \frac{b_2 s + k_2}{m b_2 s^3 + m k_2 s^2 + (k_1 + k_2) b_2 s + k_1 k_2}$$

Checks Static compliance is the reciprocal of the ground spring stiffness: the damper transmits no static force, so the series spring has no static extension.

PART 2

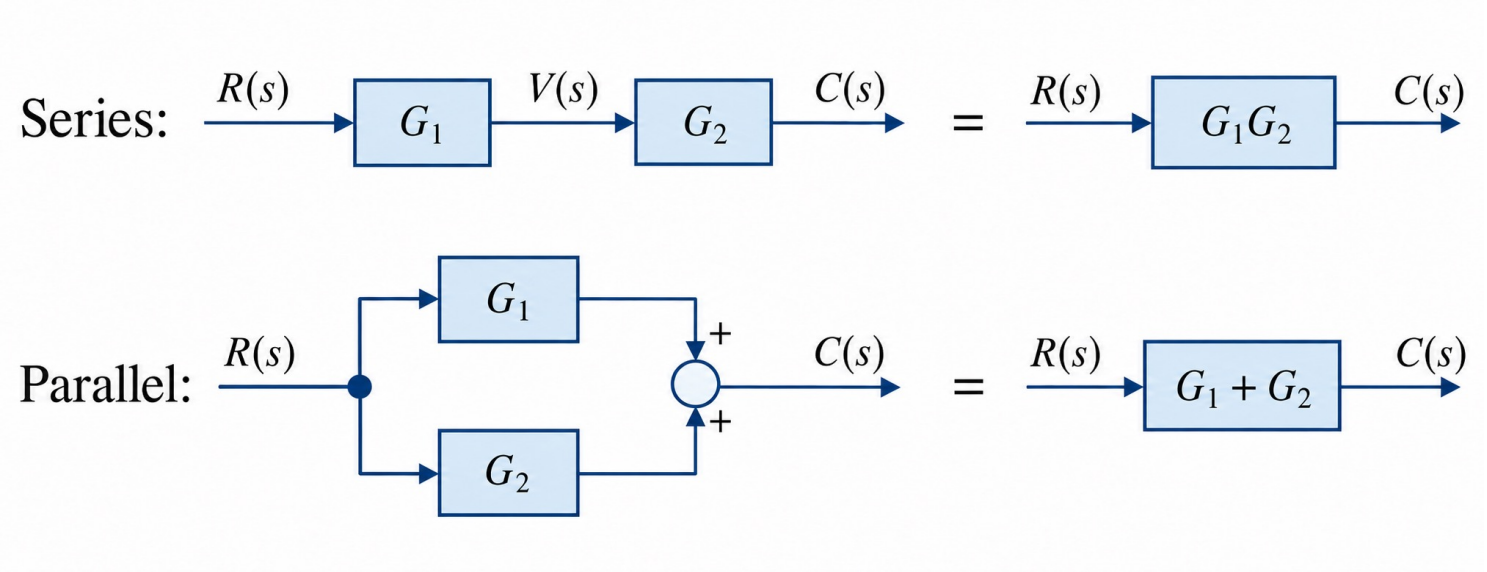
Block diagrams and feedback

01 Series, parallel and reduction rules

02 Feedback derivation and signal ratios

03 Worked reductions and disturbance inputs

Series and parallel connections



Series

$$V = G_1R, C = G_2V \Rightarrow \frac{C}{R} = G_1G_2$$

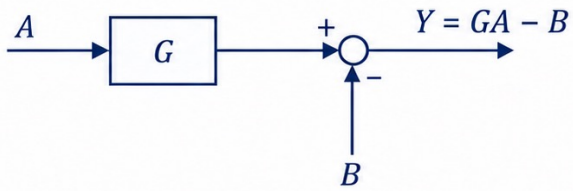
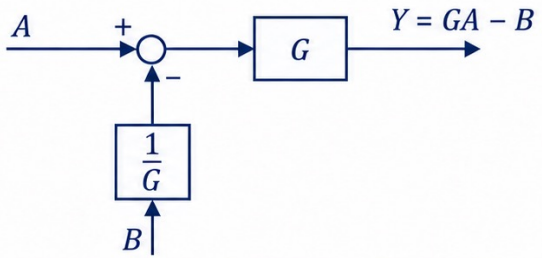
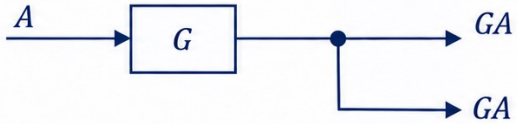
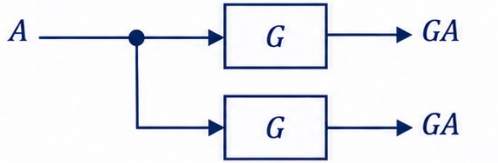
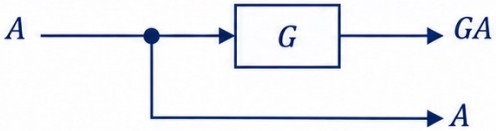
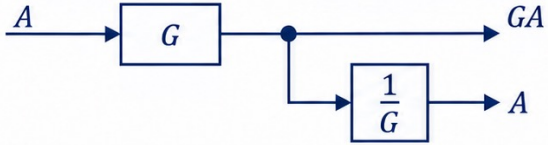
Parallel

$$C = G_1R + G_2R \Rightarrow \frac{C}{R} = G_1 + G_2$$

Block diagram reduction rules

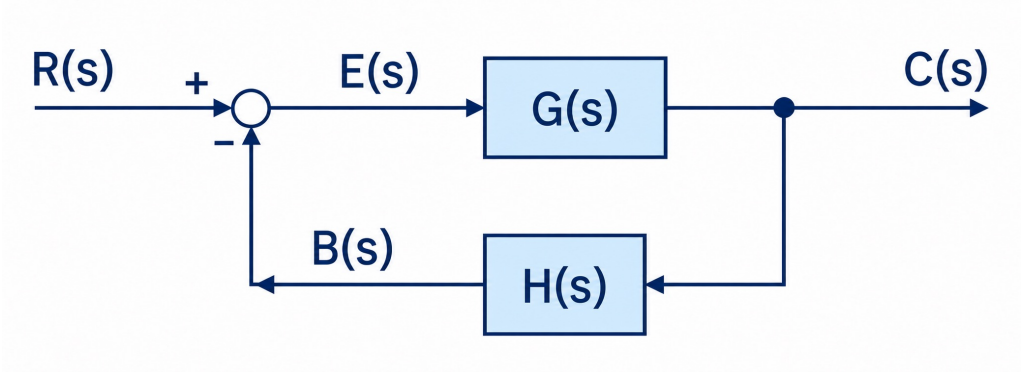
CORE

Moving a summing junction must preserve the summed signal. Moving a takeoff point must preserve every branch signal.

Rule	Original diagram	Equivalent diagram
Move summing point before G		
Move take-off before G		
Move take-off after G		

Negative feedback: the derivation

Zero initial conditions.



Signal equations

$E = R - B, \qquad B = HC, \qquad C = GE$

Substitution and rearrangement

$$E = R - HGE$$

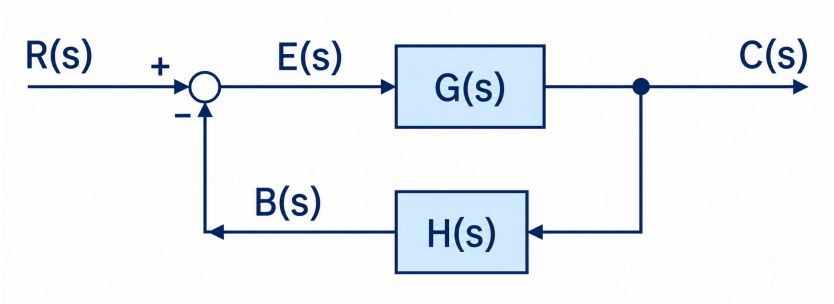
$$(1 + GH)E = R$$

$$\frac{E}{R} = \frac{1}{1 + GH}$$

$$\frac{C}{R} = G\frac{E}{R} = \frac{G}{1 + GH}$$

Forward path, open loop and closed loop

The signal ratio specifies which transfer function we mean.



Forward path

$$\frac{C}{E} = G$$

From the error signal to the output.

Open loop (OLTF)

$$\frac{B}{E} = GH = L$$

Around G and H, excluding the negative sign at the summer.

Closed loop (CLTF)

$$\frac{C}{R} = \frac{G}{1 + GH}$$

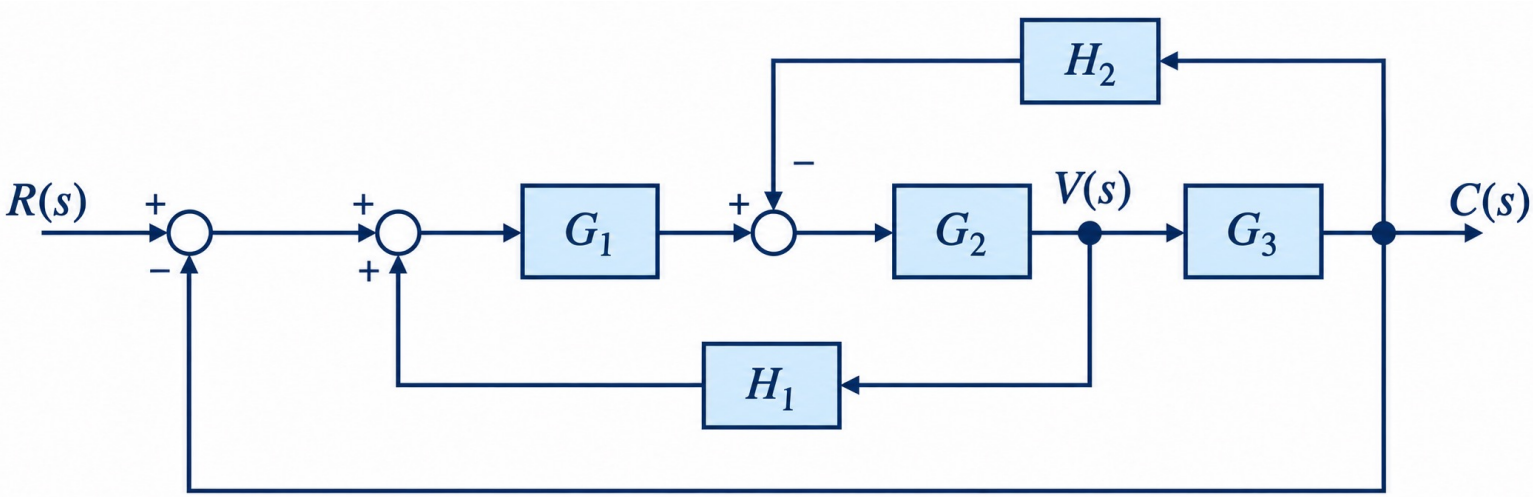
From the external reference to the output, with feedback.

With $H = 1, L = G$. Without unity feedback, G and GH are different.

Example 1: multiloop reduction

EXAMPLE

Find $C(s)/R(s)$. The H_1 branch has positive feedback.



H_2 returns C before G_2: negative.
H_1 returns V before G_1: positive.
Unity return: subtract C from R.

$C = G_3 V$

All blocks are scalar.
Initial conditions are zero.

Example 1: reduction and algebra check

EXAMPLE

1 Move the H_1 summing point after G_1

The H_1 return branch gains a factor G_1. The signal entering G_2 is:

$$G_1(R - C) + G_1H_1V - H_2G_3V, \qquad C = G_3V$$

2 Combine the inner feedback terms

$$V = G_2[G_1(R - C) - (G_3H_2 - G_1H_1)V]$$

$$\Delta = 1 + G_2G_3H_2 - G_1G_2H_1$$

$$\frac{V}{R - C} = \frac{G_1G_2}{\Delta}, \qquad \frac{C}{R - C} = \frac{G_1G_2G_3}{\Delta}$$

3 Close the outer unity negative-feedback loop

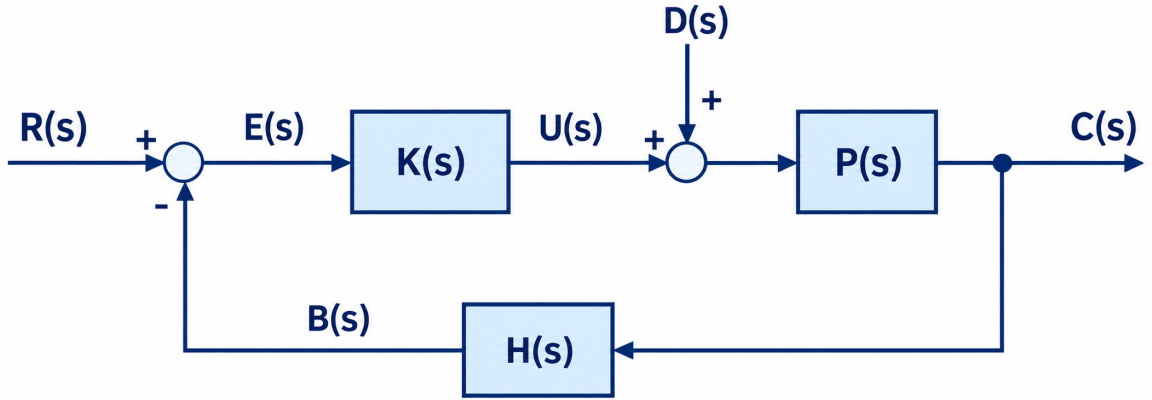
$$\frac{C}{R} = \frac{G_1G_2G_3}{1 + G_2G_3H_2 - G_1G_2H_1 + G_1G_2G_3}$$

Check: $H_1 = H_2 = 0 \Rightarrow \frac{C}{R} = \frac{G_1G_2G_3}{1 + G_1G_2G_3}$

Example 2: reference input and disturbance

EXAMPLE

Find C/R with $D = 0$, and C/D with $R = 0$. **Keep the feedback connected**.



$$C = P[KE + D], \qquad E = R - HC$$

$$C = KP(R - HC) + PD$$

$$(1 + KPH)C = KPR + PD$$

$$\frac{C}{R} = \frac{KP}{1 + KPH} \quad (D = 0) \qquad \frac{C}{D} = \frac{P}{1 + KPH} \quad (R = 0)$$

PART 3

Poles, zeros and system responses

01 Definitions and response modes

02 System poles and applied inputs

03 Zero and base-excitation examples (optional)

Poles and zeros on the s-plane

CORE

A complex root has two coordinates: $s = \sigma + j\omega, \quad j^2 = -1$

Calculation

$$G(s) = \frac{2s + 10}{s^2 + 2s + 10}$$

Poles: denominator = 0

$$s^2 + 2s + 10 = 0$$

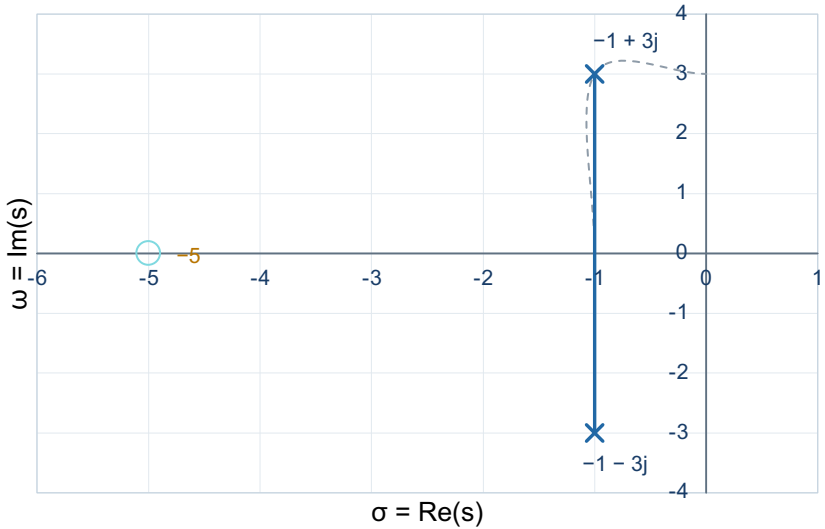
$$(s + 1)^2 = -9$$

$$p_{1,2} = -1 \pm 3j$$

Zero: numerator = 0

$$2s + 10 = 0 \Rightarrow z = -5$$

s-plane: × pole, ○ zero



At −1 + 3j: $\sigma = -1$ and $\omega = 3$ rad/s.

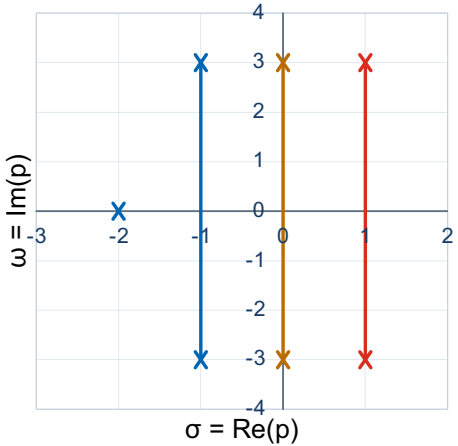
Cancel common factors first. Here, none cancel.

Pole location and time response

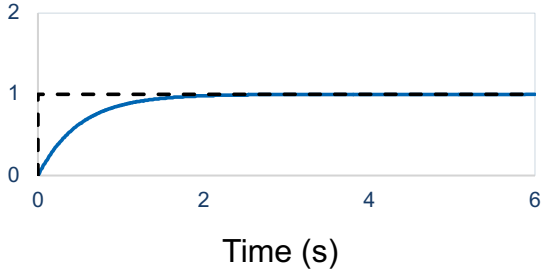
CORE

Common unit-step input. Zero initial conditions. Dashed black: input. Coloured: output.

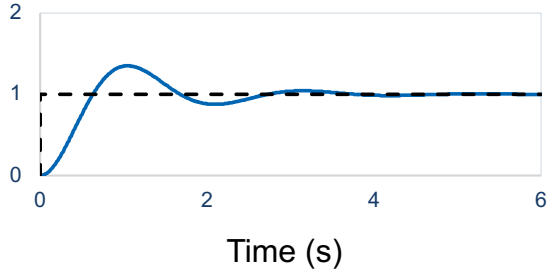
Pole locations



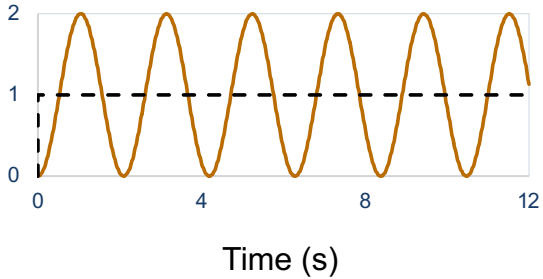
A Stable $p = -2$



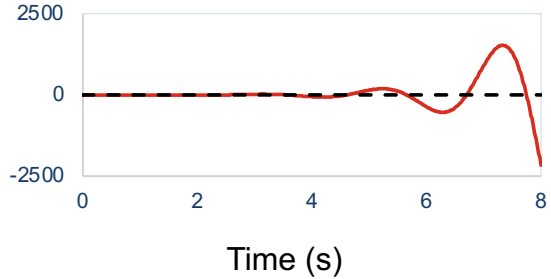
B Stable $p = -1 \pm 3j$



C Marginal $p = \pm 3j$



D Unstable $p = 1 \pm 3j$



BIBO means bounded-input bounded-output stability. C fails this test; D is unstable. Panel scales differ.

Impulse response

EXAMPLE

New fixed-base example: m=1 kg, b=10 N·s/m, k=50 N/m; zero initial state.

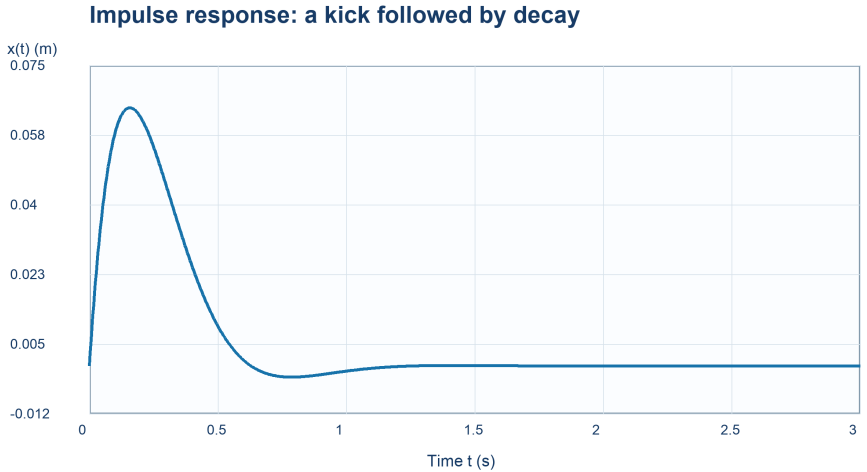
$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{1}{s^2 + 10s + 50}$$

$$X(s) = \frac{1}{s^2 + 10s + 50} \cdot 1$$

$$= \frac{1}{5} \frac{5}{(s+5)^2 + 5^2}$$

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \frac{1}{5} e^{-5t} \sin 5\,t$$

$$F(s) = 1, \qquad s^2 + 10s + 50 = (s+5)^2 + 5^2$$




```
G = tf(1,[1 10 50]);
impulse(G,0:0.01:3); grid on
```

Step response

EXAMPLE

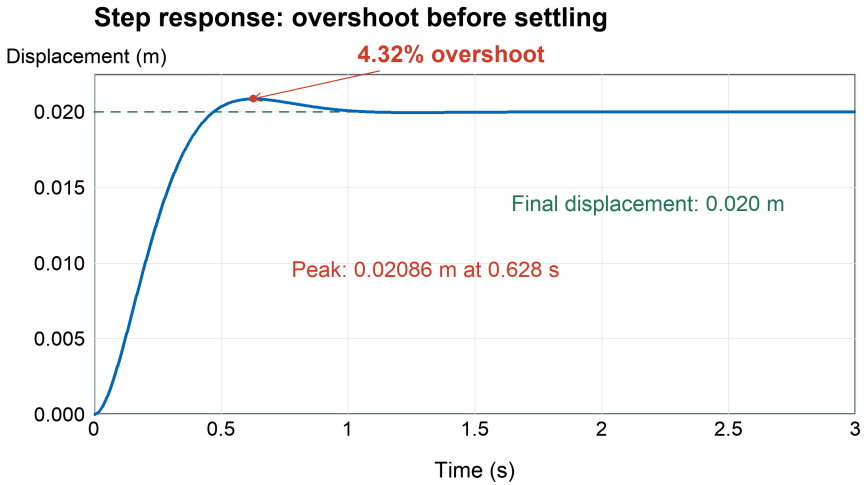
A 1 N *force stays on, so the displacement settles at* F/k

Partial fractions and inverse

$$X(s) = \frac{1}{50}\left[\frac{1}{s} - \frac{s+10}{(s+5)^2+5^2}\right]$$

$$x(t) = \frac{1}{50}\left[1 - e^{-5t}(\cos 5\,t + \sin 5\,t)\right]$$

$1/s$ term: final displacement 0.020 m. Remaining terms decay.




```
G = tf(1,[1 10 50]);
step(G,0:0.01:3); grid on
```

The step-input pole at the origin produces the constant output term.
It comes from the input and does not change the system's poles or stability.

Ramp response

EXAMPLE

A force that rises at 1 N/s *produces a rising displacement with finite lag*

Coefficient match and inverse

$$X(s) = \frac{1}{s^2 + 10s + 50}\frac{1}{s^2}$$
$$= \frac{a_1}{s^2} + \frac{a_2}{s} + \frac{a_3 s + a_4}{s^2 + 10s + 50}$$

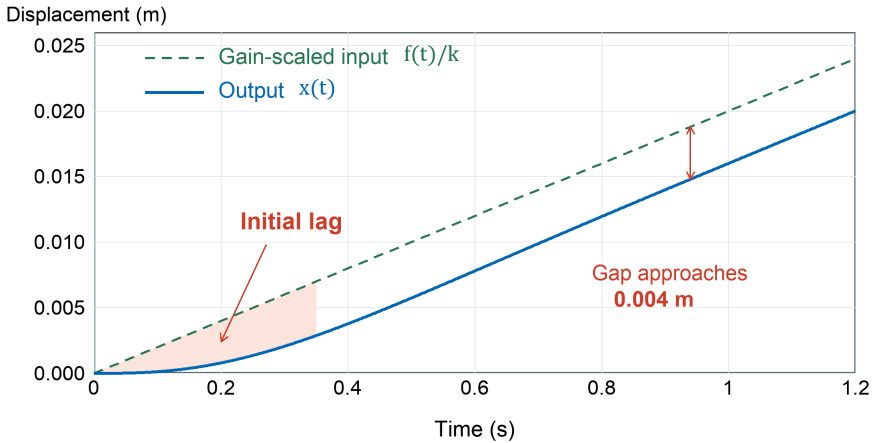
$$a_1 = \frac{1}{50}, a_2 = -\frac{1}{250}, a_3 = \frac{1}{250}, a_4 = \frac{1}{50}$$

$$x(t) = \frac{t}{50} - \frac{1}{250} + \frac{e^{-5t}\cos 5\,t}{250}$$

Check: the derivative equals the preceding unit-step response.

```
t=0:0.01:1.2; G=tf(1,[1 10 50]);
lsim(G,t,t); grid on
```

Ramp response: initial lag and continuing offset



Base excitation changes the numerator

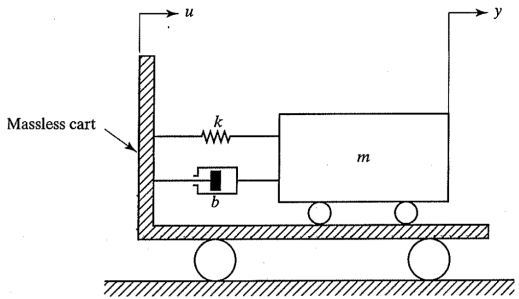
OPTIONAL

The cart moves; spring and damper forces depend on relative motion

Physical model

Input $u(t)$: cart displacement.
Output $y(t)$: mass displacement.

Given $m = 10$ kg, $b = 20$ N·s/m and $k = 100$ N/m.



Zero-state transform

$$m\ddot{y} + b(\dot{y} - \dot{u}) + k(y - u) = 0$$

$$(ms^2 + bs + k)Y(s) = (bs + k)U(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{bs + k}{ms^2 + bs + k} = \frac{2s + 10}{s^2 + 2s + 10}$$

Base motion enters through both spring and damper.

Same poles as the force path; numerator bs+k.

Right-half-plane zero: undershoot

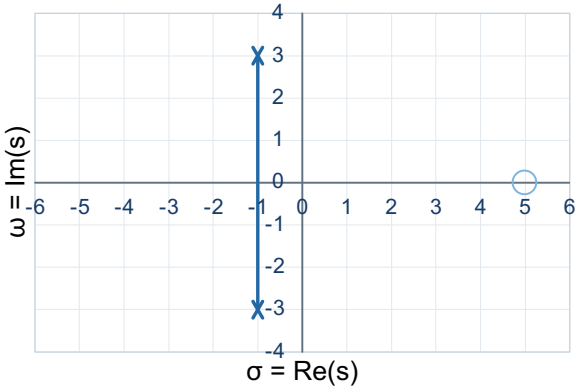
OPTIONAL

Same stable poles and final value

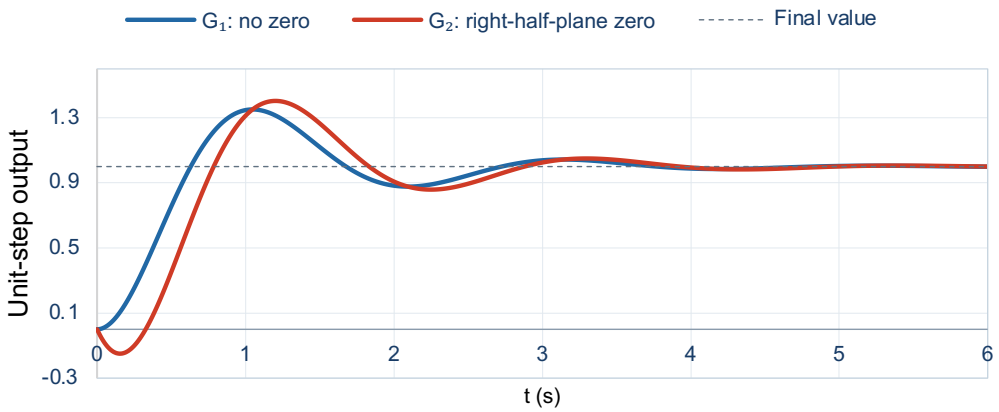
$$G_1(s) = \frac{10}{s^2 + 2s + 10}$$

$$G_2(s) = \frac{10 - 2s}{s^2 + 2s + 10}$$

No finite zero



Right-half-plane zero at +5



Shared poles: −1 ± 3j. Zero: +5.
Both outputs settle at 1; the RHP zero reverses the initial response. This illustrative model is separate from the passive base model.

Undershoot: complete step-response solution

OPTIONAL

Step input, partial fractions and inverse transform

$$Y(s) = \frac{G_2(s)}{s} = \frac{10 - 2s}{s(s^2 + 2s + 10)}$$

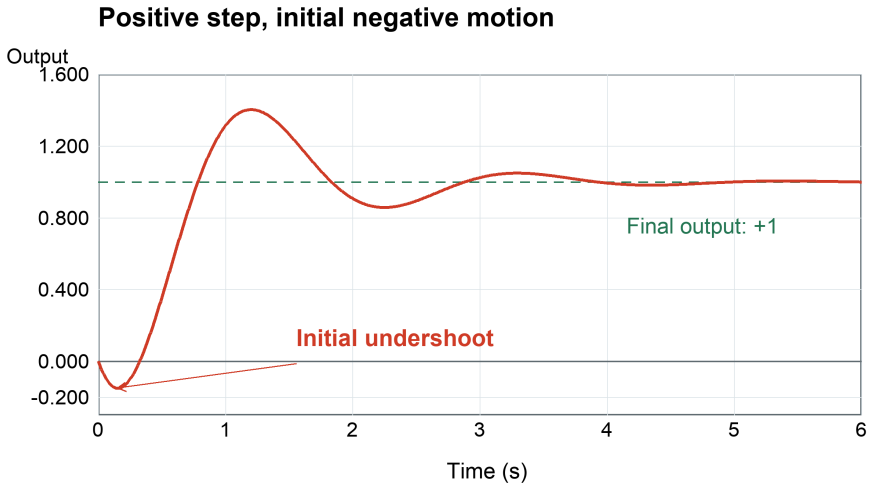
$$= \frac{1}{s} - \frac{s + 4}{s^2 + 2s + 10}$$

$$= \frac{1}{s} - \frac{s + 1}{(s + 1)^2 + 3^2} - \frac{3}{(s + 1)^2 + 3^2}$$

$$y(t) = 1 - e^{-t}\cos 3t - e^{-t}\sin 3t$$

Check: $y(0^+) = 0, \quad y(\infty) = 1$

Initial slope <0; minimum ≈−0.150; final output +1.



MATLAB commands for connected systems

OPTIONAL

Responses

```
sys = tf(num,den)
impulse(sys,t)
step(sys,t)
lsim(sys,u,t)
```

Interconnections and partial fractions

```
series(sys1,sys2)
parallel(sys1,sys2)
feedback(sysg,sysh)
[r,p,k] = residue(num,den)
```

Label time and units; check poles, final value and the hand solution.


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